

Ca DOWN

PTH LOW / NORMAL

PTH HIGH

PTH SHOULD RISE

INAPPROPRIATE

APPROPRIATE / LOOK ELSEWHERE

Ca²⁺

Ca DOWN

IONIZED

ALKALOSIS TRAP

TOTAL Ca NORMAL, IONIZED LOW

Ca²⁺

Ca DOWN

CHVOSTEK

LONG QT

TROUSSEAU MORE SPECIFIC

SPASM

CLASSIC-SIGNS

DO NOT MIX

Ca GLUCONATE IV

Ca CHLORIDE

ACUTE TREATMENT

REPLACE Mg FIRST

PO₄ HIGH

NO PTH PHOSPHATURIA

REFRACTORY

OB / RENAL FAILURE

DIGEORGE 22q11

AUTOIMMUNE APS

POSTSURGICAL

Ca WON'T CORRECT UNTIL Mg FIXED

Mg LOW

Mg HIGH

CKD

RECEPTOR BROKEN

SALCIUM SOAP

SAPONIFICATION

PO₄ VERY HIGH

PO₄ LOW

TUMOR LYSIS

RHABDO

VIT D DEF

SUN

25-D DOWN

CKD

PO₄ DOWN

ALP UP

PO₄ HIGH

CALCITRIOL DOWN

PTH RESISTANCE

PANCREATITIS

CHVOSTEK

LONG QT

TROUSSEAU MORE SPECIFIC

SPASM

CLASSIC-SIGNS

DO NOT MIX

Ca GLUCONATE IV

Ca CHLORIDE

ACUTE TREATMENT

REPLACE Mg FIRST

PO₄ HIGH

NO PTH PHOSPHATURIA

REFRACTORY

OB / RENAL FAILURE

DIGEORGE 22q11

AUTOIMMUNE APS

POSTSURGICAL

Ca WON'T CORRECT UNTIL Mg FIXED

Mg LOW

Mg HIGH

CKD

RECEPTOR BROKEN

SALCIUM SOAP

SAPONIFICATION

PO₄ VERY HIGH

PO₄ LOW

TUMOR LYSIS

RHABDO

VIT D DEF

SUN

25-D DOWN

CKD

PO₄ DOWN

ALP UP

PO₄ HIGH

CALCITRIOL DOWN

PTH RESISTANCE

PANCREATITIS

CHVOSTEK

LONG QT

TROUSSEAU MORE SPECIFIC

SPASM

CLASSIC-SIGNS

1. Ca DOWN / PTH should rise (judge)

WHAT YOU SEE

Top: 'Ca DOWN' with a judge labeled 'PTH SHOULD RISE'.

WHAT IT ENCODES

In true hypocalcemia, the CaSR should drive PTH UP (appropriate compensation). The diagnostic fork hangs on the PTH level: low/inappropriately-normal PTH points to hypoparathyroidism (gland failure), whereas high PTH is an appropriate secondary response pointing elsewhere (vitamin D deficiency, CKD, PTH resistance, sequestration). Always confirm hypocalcemia is real first.

BOARD TRAP

A hypocalcemic patient with a 'normal' PTH is read as 'PTH is fine, so the parathyroid is not the problem' — WRONG. During hypocalcemia PTH SHOULD be high; an inappropriately normal PTH is actually inappropriate and indicates hypoparathyroidism. The discriminator: interpret PTH relative to the calcium — normal PTH with low Ca is pathologic, not reassuring.

2. PTH low/normal = inappropriate (left)

WHAT YOU SEE

Left arrow: 'PTH LOW/NORMAL — INAPPROPRIATE'.

WHAT IT ENCODES

Low or inappropriately normal PTH in the setting of hypocalcemia signals HYPOPARATHYROIDISM — the gland cannot mount the expected response. Causes include post-surgical (most common), autoimmune (APS-1), hypomagnesemia (functional), infiltrative disease, and genetic (DiGeorge/22q11). Chemistry: low Ca, HIGH phosphate (no PTH-driven phosphaturia), low calcitriol.

BOARD TRAP

A post-thyroidectomy patient with low calcium and low PTH is worked up for vitamin D deficiency — WRONG/inefficient. Low PTH with low Ca is hypoparathyroidism (here surgical), characterized by HIGH phosphate, whereas vitamin D deficiency shows HIGH PTH and LOW phosphate. The discriminator: PTH direction plus phosphate splits hypoPTH (low PTH, high PO4) from vit D deficiency (high PTH, low PO4).

3. PTH high = appropriate, look elsewhere (right)

WHAT YOU SEE

Right arrow: 'PTH HIGH — APPROPRIATE / LOOK ELSEWHERE'.

WHAT IT ENCODES

High PTH with low calcium is the parathyroid responding CORRECTLY (appropriate secondary hyperparathyroidism). Look for the underlying driver: vitamin D deficiency, chronic kidney disease, PTH end-organ resistance (pseudohypoparathyroidism), or calcium sequestration (pancreatitis, tumor lysis, rhabdo, citrate). The gland is working; find what is consuming or under-absorbing calcium.

BOARD TRAP

A patient with low Ca and high PTH is labeled 'hyperparathyroidism' and worked up for an adenoma — WRONG. PRIMARY hyperparathyroidism has HIGH calcium; high PTH WITH LOW calcium is SECONDARY (appropriate compensation). The discriminator: the calcium tells you the type — high Ca + high PTH = primary; low Ca + high PTH = secondary, so hunt the cause.

4. Postsurgical hypoparathyroidism (most common)

WHAT YOU SEE

A surgeon/thyroid: 'MOST COMMON — POSTSURGICAL'.

WHAT IT ENCODES

Post-thyroidectomy or parathyroid/neck surgery is the MOST COMMON cause of acquired hypoparathyroidism, from inadvertent removal or devascularization of the parathyroid glands. It may be transient (days-weeks) or permanent. Watch for symptomatic hypocalcemia (perioral numbness, Chvostek/Trousseau) in the first 24-72h post-op; treat with calcium and calcitriol.

BOARD TRAP

A patient 2 days after total thyroidectomy with perioral tingling has it attributed to anxiety/anesthesia — WRONG and dangerous. Post-thyroidectomy hypocalcemia from hypoparathyroidism is the leading cause and can progress to laryngospasm/tetany. The discriminator: any neuromuscular symptom post-neck-surgery is hypocalcemia until proven otherwise — check calcium urgently.

5. Autoimmune / APS hypoparathyroidism

WHAT YOU SEE

Box: 'AUTOIMMUNE / APS'.

WHAT IT ENCODES

Autoimmune destruction of the parathyroids causes hypoparathyroidism, classically as part of Autoimmune Polyendocrine Syndrome type 1 (APS-1 / APECED, AIRE gene): the triad of chronic mucocutaneous candidiasis, hypoparathyroidism, and primary adrenal insufficiency (Addison). Isolated autoimmune hypoparathyroidism also occurs. Chemistry mirrors other hypoPTH: low Ca, high PO4, low/normal PTH.

BOARD TRAP

A young patient with hypoparathyroidism and oral thrush is treated for thrush alone without screening for adrenal insufficiency — WRONG/incomplete. The candidiasis + hypoparathyroidism combination should trigger evaluation for APS-1 and concurrent Addison disease, which can be life-threatening. The discriminator: hypoPTH + candidiasis = think APS-1, screen for adrenal insufficiency.

6. Hypomagnesemia blocks PTH

WHAT YOU SEE

Box: 'Mg LOW → Mg LOW; refractory; renal failure; digoxin 22q11'.

WHAT IT ENCODES

Hypomagnesemia causes FUNCTIONAL hypoparathyroidism by two mechanisms: it impairs PTH SECRETION and induces PTH RESISTANCE at target tissues. The result is hypocalcemia with low/inappropriately-normal PTH that will NOT correct until magnesium is replaced. Common causes: alcoholism, diarrhea, diuretics, proton-pump inhibitors, aminoglycosides. Always check Mg in unexplained or refractory hypocalcemia.

BOARD TRAP

A hypocalcemic patient gets repeated IV calcium boluses with no improvement, and more calcium is ordered — WRONG. Refractory hypocalcemia that won't budge is the classic sign of unrecognized HYPOMAGNESEMIA, which blocks both PTH release and action. The discriminator: in refractory hypocalcemia, check and replace magnesium FIRST; calcium won't rise until Mg is corrected.

7. Hyperphosphatemia / PO4 high

WHAT YOU SEE

Box: 'PO4 HIGH; no PTH phosphaturia'.

WHAT IT ENCODES

In hypoparathyroidism, loss of PTH-driven renal phosphate excretion (phosphaturia) causes phosphate RETENTION, so the pattern is low calcium WITH high phosphate. This phosphate-calcium relationship is a key discriminator: hypoPTH and pseudohypoPTH both show high PO4, whereas vitamin D deficiency (high PTH causing phosphaturia) shows low/normal PO4.

BOARD TRAP

A hypocalcemic patient with high phosphate is presumed to have vitamin D deficiency — WRONG. Low Ca + HIGH PO4 points to a PTH problem (hypoparathyroidism if PTH low, pseudohypoparathyroidism if PTH high), NOT vitamin D deficiency, which gives low Ca + LOW PO4. The discriminator: phosphate splits the differential — high PO4 = PTH axis problem, low PO4 = vitamin D problem.

8. Vitamin D deficiency

WHAT YOU SEE

A dim sun: '25-OH VIT D DOWN; PO4 LOW; calcitriol down'.

WHAT IT ENCODES

Vitamin D deficiency (low 25-OH vitamin D from low sun/intake, malabsorption, or liver disease) reduces intestinal calcium AND phosphate absorption, lowering serum calcium. The low calcium drives a compensatory rise in PTH (secondary hyperparathyroidism), which causes renal phosphate wasting — hence the pattern: low Ca, LOW phosphate, HIGH PTH, low calcitriol. Treat with vitamin D (and calcium) repletion.

BOARD TRAP

A hypocalcemic patient with low phosphate and high PTH is mislabeled hypoparathyroidism — WRONG; that pattern (low PO4, high PTH) is vitamin D deficiency, the mirror image of hypoPTH (high PO4, low PTH). The discriminator: high PTH + low phosphate = vitamin D deficiency; low PTH + high phosphate = hypoparathyroidism.

9. CKD / secondary HPT

WHAT YOU SEE

A worn kidney figure at the right-hand bench labelled 'CKD — PTH UP, calcitriol DOWN, PO4 retained' — the failing kidney visually driving secondary hyperparathyroidism.

WHAT IT ENCODES

In CKD, the failing kidney loses 1-alpha-hydroxylase activity (low calcitriol → less gut calcium absorption) AND cannot excrete phosphate (hyperphosphatemia, which further binds calcium and suppresses calcitriol). The combined hypocalcemia drives secondary hyperparathyroidism: high PTH, low calcitriol, high phosphate. Chronic high PTH can progress to tertiary HPT (autonomous, hypercalcemic).

BOARD TRAP

A CKD patient with hypocalcemia, high phosphate, and high PTH is treated only with calcium without addressing phosphate or calcitriol — WRONG/incomplete. CKD-mineral bone disease needs phosphate binders, active vitamin D (calcitriol), and sometimes calcimimetics. The discriminator: CKD hypocalcemia is multifactorial (low calcitriol + phosphate retention) — manage the whole axis, not just calcium.

10. Pseudohypoparathyroidism (PHP, resistance)

WHAT YOU SEE

A figure with a 'broken receptor' icon labelled 'RECEPTOR BROKEN — PHP, PTH RESISTANCE' — the snapped receptor is the bridge to end-organ resistance: PTH is shouting but the target can't hear it.

WHAT IT ENCODES

Pseudohypoparathyroidism (PHP type 1a, GNAS mutation, maternal allele) is end-organ RESISTANCE to PTH: the gland works and secretes plenty of PTH, but the kidney/bone receptor (Gs-alpha coupled) cannot respond. Result: low Ca, high PO4, but HIGH PTH. Type 1a has Albright hereditary osteodystrophy — short stature, round face, brachydactyly (short 4th/5th metacarpals), obesity.

BOARD TRAP

A short patient with round face, short fourth metacarpals, low Ca, and high phosphate is diagnosed with true hypoparathyroidism — WRONG. The HIGH PTH (not low) plus the Albright phenotype defines pseudohypoparathyroidism (PTH resistance). The discriminator: identical chemistry to hypoPTH (low Ca, high PO4) but PTH is HIGH — that elevated PTH is the key distinguishing feature.

11. Sequestration: pancreatitis / saponification

WHAT YOU SEE

A panel labelled 'CALCIUM SOAP — SAPONIFICATION, PANCREATITIS, RHABDO, TUMOR LYSIS' — the 'soap' image is the bridge: free fatty acids chelate calcium into soaps, locking it away.

WHAT IT ENCODES

Calcium can be acutely consumed or sequestered: in acute pancreatitis, free fatty acids from fat necrosis chelate calcium (saponification → calcium soaps); in tumor lysis syndrome and rhabdomyolysis, released phosphate complexes with calcium; massive transfusion citrate chelates ionized calcium. PTH rises appropriately, but the calcium is being bound/lost faster than it can be replaced.

BOARD TRAP

A patient with acute pancreatitis and hypocalcemia has the low calcium dismissed as incidental — WRONG; hypocalcemia in pancreatitis (from saponification) is a marker of SEVERITY and a Ranson/prognostic criterion. The discriminator: low calcium in pancreatitis is clinically meaningful, signals fat necrosis and worse prognosis, and may need repletion if symptomatic.

12. Alkalosis trap (total normal, ionized low)

WHAT YOU SEE

Box: 'ALKALOSIS TRAP — TOTAL Ca NORMAL, IONIZED LOW'.

WHAT IT ENCODES

Alkalosis increases calcium binding to albumin, lowering the active IONIZED fraction while leaving TOTAL calcium normal. This produces full hypocalcemic symptoms (tingling, tetany, Chvostek/Trousseau) with a reassuring-looking total calcium and total-body calcium that is actually normal. The classic acute cause is hyperventilation-induced respiratory alkalosis.

BOARD TRAP

A hyperventilating patient with tetany and a normal total calcium is told they have no calcium problem — WRONG. Alkalosis has dropped the ionized fraction despite a normal total. The discriminator: when symptoms suggest hypocalcemia but total Ca is normal, check pH and ionized calcium; treat the alkalosis rather than infusing calcium.

13. Corrected-Ca formula

WHAT YOU SEE

Box: 'CORRECTED Ca = $\text{Ca} + 0.8 \times (4 - \text{albumin})$ '.

WHAT IT ENCODES

Before diagnosing true hypocalcemia, correct total calcium for albumin: corrected Ca = measured Ca + 0.8 x (4 - albumin in g/dL), or measure ionized calcium directly. This rules out pseudohypocalcemia from hypoalbuminemia (common in hospitalized, cirrhotic, nephrotic, and critically ill patients), avoiding unnecessary calcium treatment.

BOARD TRAP

A hospitalized patient with low albumin and a low total calcium is launched into a full hypocalcemia workup (PTH, vitamin D, magnesium) without correcting first — WRONG/wasteful. The corrected value may be normal (pseudohypocalcemia). The discriminator: always albumin-correct or check ionized calcium FIRST; only a truly low corrected/ionized calcium warrants the workup.

14. Acute treatment — IV calcium gluconate, replace Mg first

WHAT YOU SEE

Bottom-left: 'ACUTE TREATMENT — Ca GLUCONATE IV FIRST; REPLACE Mg FIRST'.

WHAT IT ENCODES

Symptomatic or severe hypocalcemia (tetany, seizures, laryngospasm, prolonged QT/arrhythmia) is treated with IV calcium gluconate (e.g., 1-2 g in 50-100 mL D5W over 10-20 min, then an infusion). Concurrently CHECK and REPLACE magnesium, since hypomagnesemia causes refractory hypocalcemia. For chronic hypoPTH, add oral calcium and calcitriol (active vitamin D).

BOARD TRAP

Calcium CHLORIDE is given through a peripheral IV for symptomatic hypocalcemia — WRONG choice peripherally. Calcium chloride is highly irritating and causes tissue necrosis on extravasation, so it is preferred only via central access; calcium GLUCONATE is the peripheral agent of choice. The discriminator: peripheral line = gluconate; central line = chloride; and always replace magnesium or calcium won't correct.

15. Classic signs: Chvostek / Trousseau / long QT

WHAT YOU SEE

Bottom-right: 'CLASSIC SIGNS — CHVOSTEK, TROUSSEAU, LONG QT'.

WHAT IT ENCODES

Hypocalcemia increases neuromuscular excitability (low ionized Ca lowers the threshold for nerve depolarization): Chvostek sign (tapping the facial nerve anterior to the ear → facial twitch), Trousseau sign (BP cuff inflated above systolic for 3 min → carpopedal spasm, the more specific sign), perioral/acral paresthesias, and a PROLONGED QT interval (risk of torsades de pointes), plus seizures and laryngospasm in severe cases.

BOARD TRAP

An ECG with a prolonged QT in a symptomatic patient is attributed to hypercalcemia — WRONG and reversed. Hypocalcemia PROLONGS the QT (lengthened ST segment, torsades risk); HYPERcalcemia SHORTENS it. The discriminator: low Ca = long QT, high Ca = short QT — never mix the two directions.